
Integrative Programming and Technologies III

Test 2

Total: 40 marks

17 May 2022 Time: 1 hour

Question 1-6 6 marks

Indicate whether the following statements are **TRUE** or **FALSE**.

1. Web services use XML for data exchange. (T)
2. In RMI, an object to be transferred over a network should be serialized. (T)
3. On a socket, the client's output stream connects to the server's output stream. (F)
4. The **openStream** () method of the URL object can read directly from URL. (T)
5. The **openConnection**() method of the URL object allows reading from and writing to a stream. (T)
6. CORBA is widely used to integrate legacy systems because it is platform independent. (T)

Question 7 2 marks

Briefly explain one reason why XML is preferable to database management.

Answer: Any correct reason, e.g., XML is platform independent and, this can reduce development cost (e.g., develop once, use on many platforms).

Question 8 1 mark

Which of the following statements is or are true about the StaX API?

- A. Allows both read and write
- B. Can handle document of any size
- C. Both A and B
- D. None of the above

Answer: C

Question 9 6 marks

The following shows part of XML code.

```
<?xml version="1.0" encoding="UTF-8"?>
<Course>
  <Subject code="IPRT301">
    <name>Integrated programming 1</name>
    <lecturer>Dr L Adu</lecturer>
  </Subject>
  <Subject code=" PBDE401">
    <name>Platform-based programming 1</name>
    <lecturer>FZ Bam</lecturer>
  </Subject>
</Course>
```

Write the missing parts (ONLY) of the DTD code below for the XML code. (6)

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<!ELEMENT name (#PCDATA)>
```

```
<!ELEMENT lecturer (#PCDATA)>
```

Answer:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<!ELEMENT Course (Subject)+> ✓✓
```

```
<!ATTLIST Subject code CDATA #REQUIRED>✓✓
```

```
<!ELEMENT Subject (name, lecturer)> ✓✓
```

```
<!ELEMENT name (#PCDATA)>
```

```
<!ELEMENT lecturer (#PCDATA)>
```

Question 10 4 marks

The following is an XML code.

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<Course>
```

```
  <Subject code="IPRT301">
```

```
    <name>Integrated programming 1</name>
```

```
    <lecturer>Dr L Adu</lecturer>
```

```
  </Subject>
```

```
  <Subject code="PBDE401">
```

```
    <name>Platform-based programming 1</name>
```

```
    <lecturer>FZ Bam</lecturer>
```

```
  </Subject>
```

```
</Course>
```

Write the missing parts (ONLY) of the XSL code below to get the values in the XML document. (4)

```
<xsl:for-each select="_____">
```

```
  <tr>
```

```
    <td><xsl:value-of select="_____" /></td>
```

```
    <td><xsl:value-of select="_____" /></td>
```

```
    <td><xsl:value-of select="_____" /></td>
```

```
  </tr>
```

```
</xsl:for-each>
```

Answer:

```
<xsl:for-each select="Course/Subject" ✓✓>
```

```
  <tr>
```

```
    <td><xsl:value-of select="@code" ✓✓ /></td>
```

```
    <td><xsl:value-of select="name" ✓✓ 1/2 /></td>
```

```
    <td><xsl:value-of select="lecturer" ✓✓ 1/2 /></td>
```

```
  </tr>
```

```
</xsl:for-each>
```

Question 11 3 marks

Match the following technologies to the correct distributed architecture

1	RMI	A	Distributed object architecture
2	CORBA	B	Web service architecture
3	SOAP	C	Micro service architecture
			None of these

1. A; 2. A; 3. B

Question 12 2 mark

A client transfers an object of a class **Mark** on a network in an RMI application. Complete the class header below for the class Mark. (2)

```
public class Mark implements Serializable ✓✓{  
    ...  
    ...  
}
```

Question 13 17 marks

The figure below shows the class of a remote object. The method **calcFinalMark** takes two parameters (dp and exam mark) and returns the final mark as 40% of the dp and 60% of the exam mark. Use the information to answer the following questions.

```
MarkCalculatorImp  
MarkCalculatorImp()  
calcFinalMark(int, int) : int
```

1. Complete the code below for the remote interface of the remote object. (4)

```
public interface IMarkCalculator ✓ extends Remote ✓ {  
    public int calcFinalMark(int dpPar, int examPar) ✓ throws Exception ✓;  
}
```

2. Complete the code below for the remote object. (9)

```
public class MarkCalculatorImp extends UnicastRemoteObject ✓ implements  
IMarkCalculator ✓ {  
    protected MarkCalculatorImp() throws Exception ✓ { ✓  
    }  
    public int calcFinalMark(int dpPar, int examPar) throws Exception ✓ { ✓  
        return (int) Math.round((dpPar*0.4)+(0.6*examPar)); ✓ ✓  
    }  
}
```

3. Complete the client program below which connects to the remote object on a local host on port 1999. Assume the remote object is named mark in a registry. (4)

```
public class TestClient {  
    public static void main(String[] args) throws Exception{  
        //Create stub  
        IMarkCalculator  
i=(IMarkCalculator)Naming.lookup("rmi://localhost:1999/mark");✓✓✓  
        //Access method of the remote object to display the final mark  
        System.out.println(m);  
        System.out.println("Final mark: "+i.calcFinalMark(50, 50) ✓);  
    }  
}
```

Question 14 (2 marks)

Give one reason why exception handling is an important feature of distributed applications.

Any correct answer: e.g., in distributed applications, communication occurs over networks. Exception handling is important to avoid abnormal flow programs, e.g., because the network is unavailable✓✓